

Offline Coefficient-Space Comparison for ANN Maps

$n_s \in \{131, 141, 151\}$

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Objective

This report compares previously trained maps in pure offline reconstruction mode:

$$(\mu, t) \mapsto q_s(\mu, t),$$

without running online nonlinear ROM solves.

The compared models use

$$n_s \in \{131, 141, 151\},$$

using the same linear PROM reference trajectories.

The row labeled Old DD-ANN corresponds to the previously trained data-driven ANN checkpoint from the original 250x250 study; it is imported here only as a fixed baseline for comparison.

Reference trajectories are the linear PROM coefficients $q_N^{lin}(t; \mu)$ with $n_{tot} = 151$ at

$$\mu^{(v)} = (4.875, 0.0225), \quad \mu^{(1)} = (4.560, 0.0190), \quad \mu^{(2)} = (5.190, 0.0260).$$

Model specifications used in this report

All maps use inputs (μ_1, μ_2, t) . Closure maps output q_s ; full data-driven maps output the complete q_N . For ANN full-coordinate maps, the hidden architecture is the same as in the training script core MLP:

(32, 64, 128, 256, 256) with ELU activations.

Table 1: ANN training setup (from Stage-3 summaries).

n_s	hidden dims	act.	drop.	batch	lr	weight decay	epochs ran	best val MSE
131	(32, 64, 128, 256, 256)	elu	0.0	128	0.001	1e-06	2000	0.1265958845615387
141	(32, 64, 128, 256, 256)	elu	0.0	128	0.001	1e-06	2000	0.13269366323947906
151	(32, 64, 128, 256, 256)	elu	0.0	128	0.001	1e-06	5000	0.23475342988967896

Error definitions

For a model with split $n_{tot} = n_p + n_s$, we compare predicted secondary coefficients against the corresponding tail of the linear reference:

$$q_s^{ref}(t; \mu) = [q_N^{lin}(t; \mu)]_{n_p+1:n_{tot}}, \quad q_s^{pred}(t; \mu) = \mathcal{M}_\theta(\mu, t).$$

For each predicted coefficient index $i = 1, \dots, n_s$:

$$E_i^{abs}(\mu) = \|q_{s,i}^{ref} - q_{s,i}^{pred}\|_2, \quad E_i^{rel}(\mu) = 100 \frac{\|q_{s,i}^{ref} - q_{s,i}^{pred}\|_2}{\|q_{s,i}^{ref}\|_2 + \varepsilon}.$$

Global secondary-trajectory error:

$$E_F(\mu) = 100 \frac{\|Q_s^{ref} - Q_s^{pred}\|_F}{\|Q_s^{ref}\|_F + \varepsilon}.$$

Since models have different n_s , each curve starts at global index $n_p + 1$ with $n_p = n_{tot} - n_s$. Therefore, missing lower global indices for some curves are expected.

Pointwise summary

Table 2: Offline coefficient errors per point (%).

Model	n_s	E_F	mean(E_i^{rel})	p95(E_i^{rel})	max(E_i^{rel})
$\mu = (4.875, 0.0225)$					
ANN	131	7.656	21.402	59.118	72.555
ANN	141	4.470	21.587	61.910	73.980
ANN	151	1.346	29.587	81.324	88.820
Old DD-ANN	151	1.904	33.610	81.629	87.530
$\mu = (4.560, 0.0190)$					
ANN	131	108.135	160.144	256.243	358.352
ANN	141	10.362	32.361	68.048	82.348
ANN	151	3.719	52.902	110.487	149.926
Old DD-ANN	151	4.139	46.352	99.670	126.930
$\mu = (5.190, 0.0260)$					
ANN	131	74.604	126.429	198.057	223.710
ANN	141	11.252	38.762	75.742	84.554
ANN	151	5.006	59.906	113.247	158.469
Old DD-ANN	151	3.489	47.900	95.019	113.932

Average over the three points

Table 3: Averaged offline coefficient errors over the three points (%).

Model	n_s	mean E_F	mean(mean(E_i^{rel}))	mean(p95(E_i^{rel}))	mean(max(E_i^{rel}))
ANN	131	63.465	102.659	171.139	218.206
ANN	141	8.695	30.903	68.567	80.294
ANN	151	3.357	47.465	101.686	132.405
Old DD-ANN	151	3.177	42.621	92.106	109.464

Per-coefficient curves

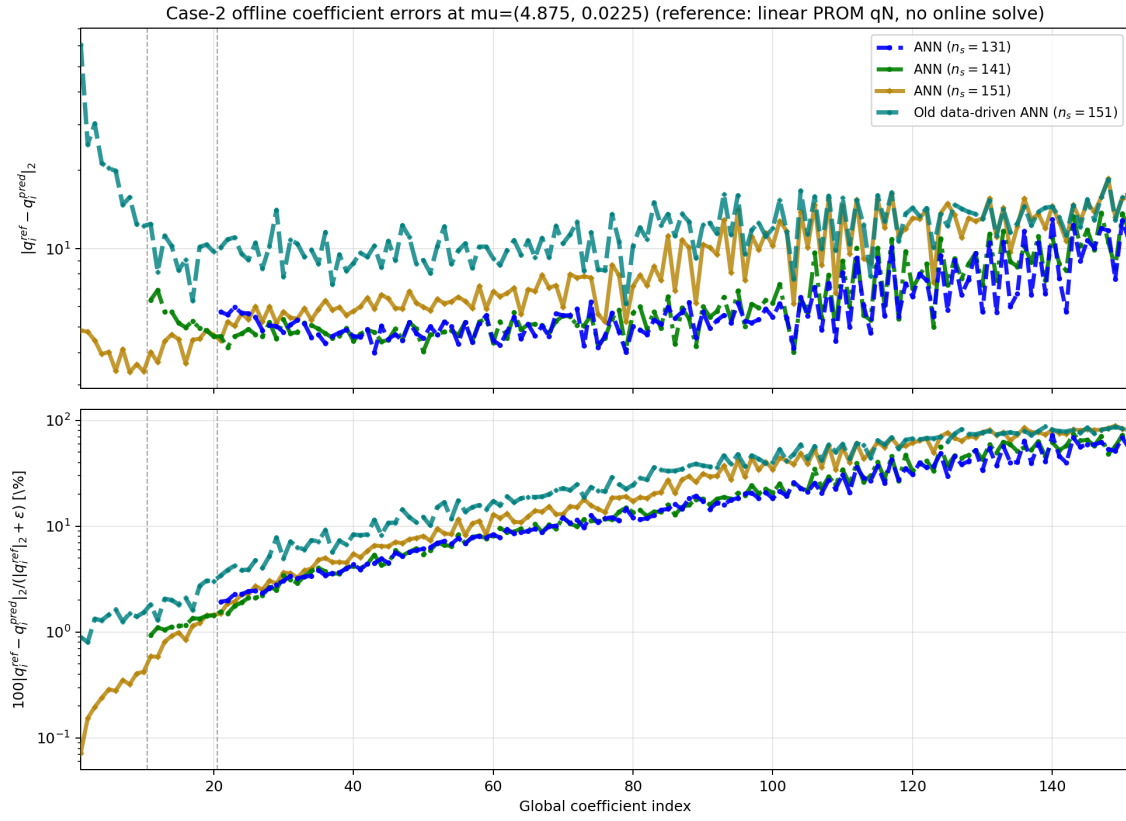


Figure 1: Absolute and relative coefficient errors vs global coefficient index at $\mu^{(v)} = (4.875, 0.0225)$.

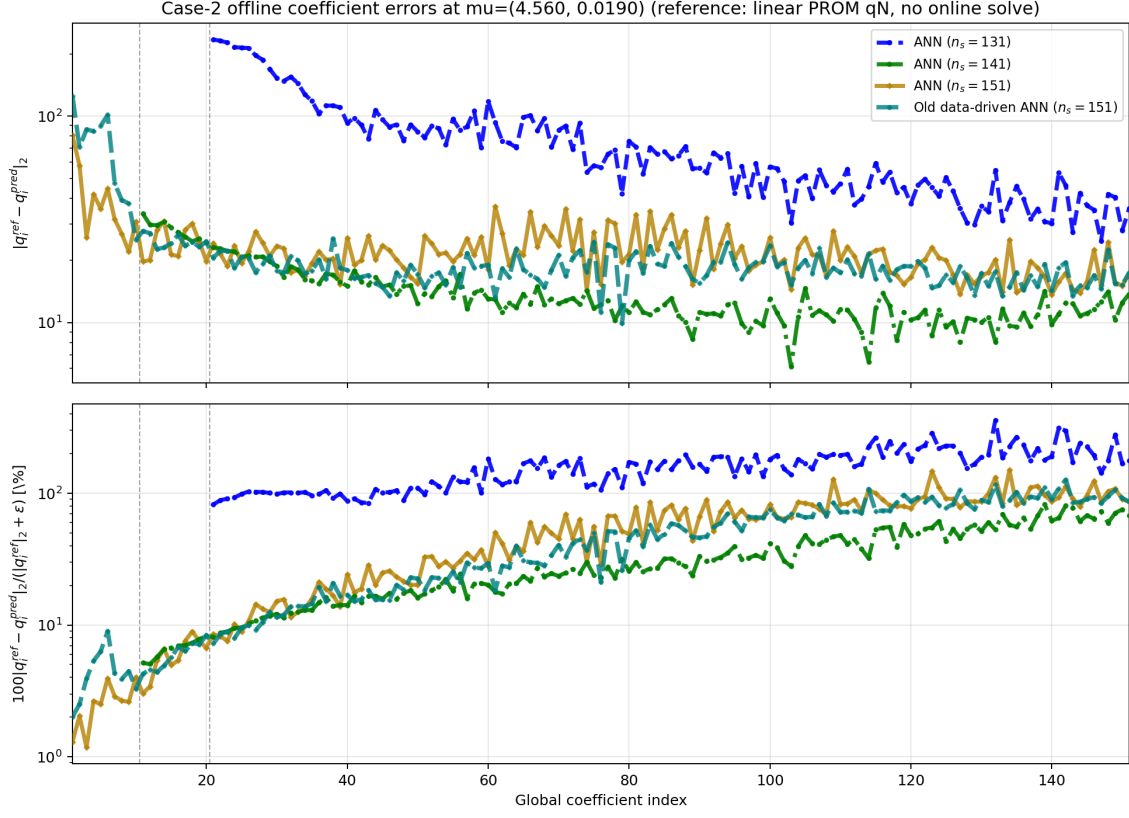


Figure 2: Absolute and relative coefficient errors vs global coefficient index at $\mu^{(1)} = (4.560, 0.0190)$.

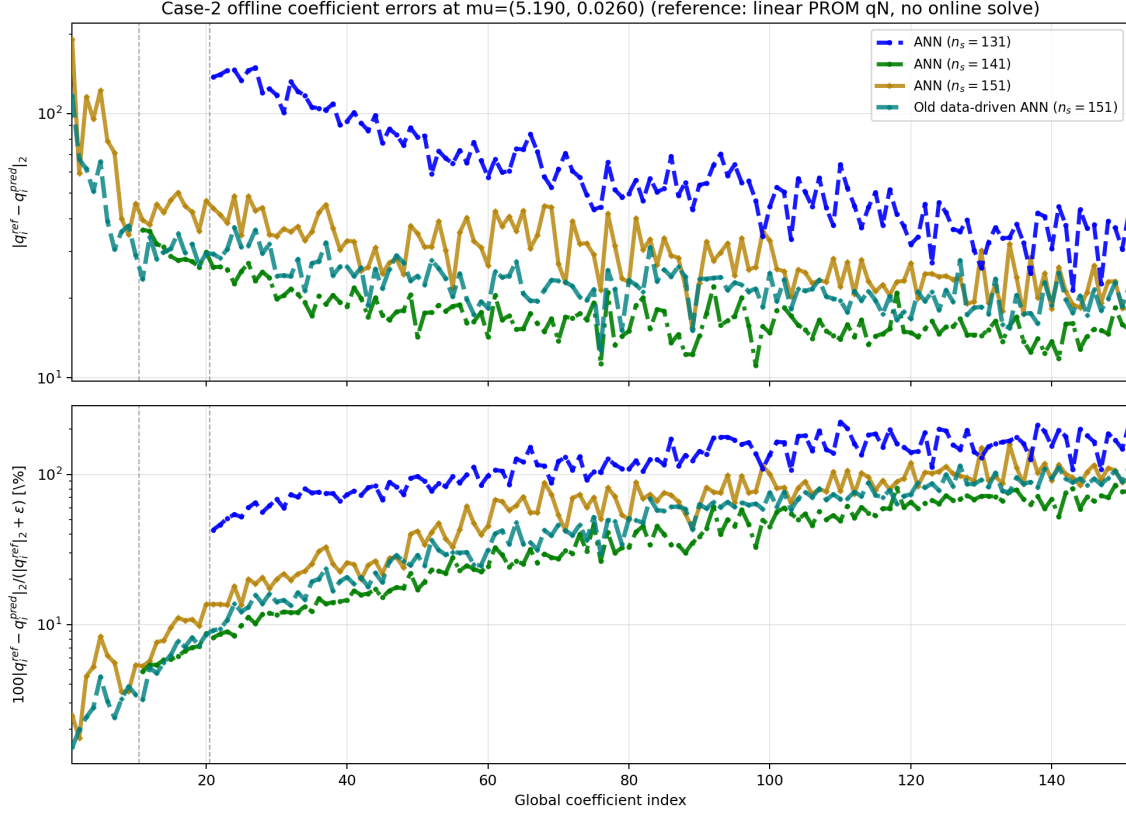


Figure 3: Absolute and relative coefficient errors vs global coefficient index at $\mu^{(2)} = (5.190, 0.0260)$.

Hybrid state-space overlays (HDM comparison)

Using the same predicted coefficients, hybrid full reduced trajectories were built as

$$q_N^{hyb} = \begin{bmatrix} q_p^{lin} \\ q_s^{pred} \end{bmatrix},$$

with $(n_p, n_s) = (20, 131)$, $(10, 141)$, and $(0, 151)$ depending on the model. Then states were reconstructed by

$$u^{hyb}(t) = u_{ref} + V_{tot} q_N^{hyb}(t),$$

and compared directly against HDM snapshots.

Table 4: Hybrid state errors per point (%).

Model	n_s	err. vs HDM (%)	err. vs linear state (%)
$\mu = (4.875, 0.0225)$			
ANN	131	0.518	0.372
ANN	141	0.534	0.403
ANN	151	0.659	0.578
Old DD-ANN	151	0.874	0.818
$\mu = (4.560, 0.0190)$			
ANN	131	5.561	5.508
ANN	141	1.049	0.963
ANN	151	1.651	1.616
Old DD-ANN	151	1.832	1.798
$\mu = (5.190, 0.0260)$			
ANN	131	3.697	3.672
ANN	141	1.080	1.007
ANN	151	2.233	2.199
Old DD-ANN	151	1.580	1.533

Table 5: Average hybrid state errors over the three points (%).

Model	n_s	mean err. vs HDM (%)	mean err. vs linear state (%)
ANN	131	3.259	3.184
ANN	141	0.888	0.791
ANN	151	1.514	1.464
Old DD-ANN	151	1.429	1.383

Offline hybrid state overlays: HDM vs ANN/GPR hybrid qN reconstructions

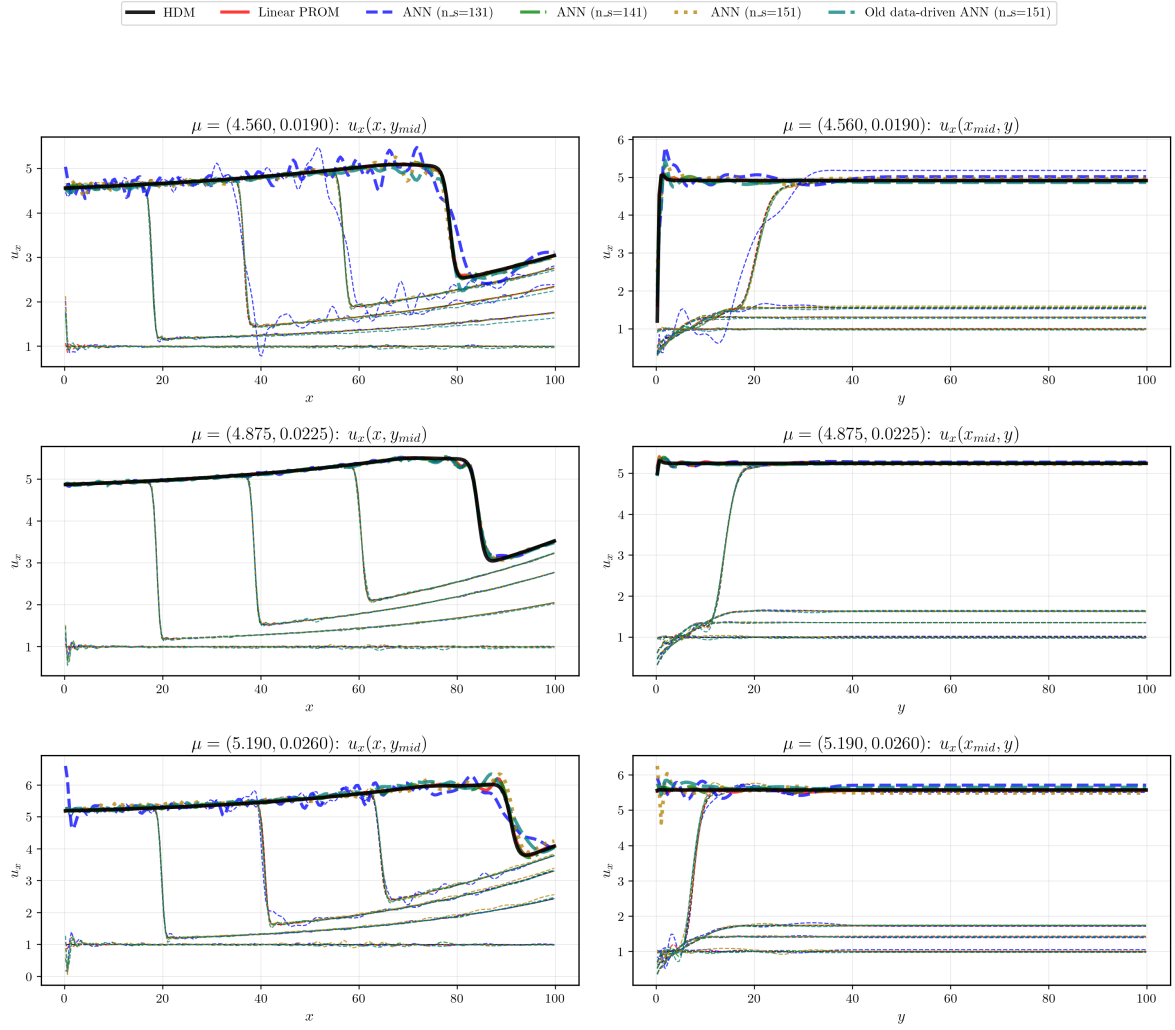


Figure 4: Hybrid offline state overlays: HDM, linear PROM reference, and learned-map hybrid reconstructions for $n_s \in \{131, 141, 151\}$.

Remarks

This report is strictly an offline map-comparison diagnostic in coefficient space. It does not include online ROM reconstruction of state trajectories.